

YULIN LUO

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👤 RESEARCH PROFILE

Ph.D. student in Computer Science at Peking University, advised by Prof. Shanghang Zhang. My research focuses on generalizable embodied foundation models for open-world robot intelligence, with emphasis on embodied evaluation, model-data co-evolution, and robust multimodal learning.

🎓 EDUCATION

Peking University, Ph.D. in Computer Science Sep. 2023 – Present

- Advisor: Prof. Shanghang Zhang.

Shanghai Jiao Tong University, B.Eng. in Automation Sep. 2019 – Jun. 2023

- GPA: 3.87/4.30; Major rank: 2/89.

🔬 RESEARCH INTERESTS

- Generalizable embodied foundation models for open-world robot intelligence.
- Embodied-agent benchmarks, multimodal evaluation, and robot-scenario data construction.
- Model-data co-evolution, visual-language-action models, and robust vision/data-centric learning.

📄 SELECTED PUBLICATIONS

- **RoboBench: A Comprehensive Evaluation Benchmark for MLLMs as Embodied Brain.**
Yulin Luo*, Chun-Kai Fan*, Menghang Dong*, Jiayi Shi*, Mingkang Zhao*, Bo-Wen Zhang*, et al., Shanghang Zhang.
European Conference on Computer Vision (ECCV), 2026. First author. Project / Code / Dataset available.
- **Decomposing the Neurons: Activation Sparsity via Mixture of Experts for Continual Test-Time Adaptation.**
Rongyu Zhang*, Aosong Cheng*, Yulin Luo*, Gaole Dai, Huanrui Yang, et al., Shanghang Zhang.
AAAI Conference on Artificial Intelligence (AAAI), 2026. Oral. Co-first author.
- **LLM as Dataset Analyst: Subpopulation Structure Discovery with Large Language Model.**
Yulin Luo*, Ruichuan An*, Bocheng Zou, Yiming Tang, Jiaming Liu, Shanghang Zhang.
European Conference on Computer Vision (ECCV), 2024. First author. Project / Code available.
- **RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization.**
Yiqing Shen*, Yulin Luo*, Dinggang Shen, Jing Ke.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022. Co-first author.
- **Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models.**
Yulin Luo*, Hao Chen*, Zhixuan Wu*, Bo Sui*, Jiaming Liu*, et al., Shanghang Zhang.
arXiv preprint, 2026. Co-first author.
- **RoboMIND: Benchmark on Multi-Embodiment Intelligence Normative Data for Robot Manipulation.**
Kun Wu, Chengkai Hou, Jiaming Liu, et al., Yulin Luo, et al.
Robotics: Science and Systems (RSS), 2025.
- **SEEA-R1: Tree-Structured Reinforcement Fine-Tuning for Self-Evolving Embodied Agents.**
Wanxin Tian, Shijie Zhang, Kun Zhang, Xiaowei Chi, Yulin Luo, et al.
Advances in Neural Information Processing Systems (NeurIPS), 2025.
- **Draw-and-Understand: Leveraging Visual Prompts to Enable MLLMs to Comprehend What You Want.**
Weifeng Lin, Xinyu Wei, Ruichuan An, Peng Gao, Bocheng Zou, Yulin Luo, et al.
International Conference on Learning Representations (ICLR), 2025.

- **Efficient Deweather Mixture-of-Experts with Uncertainty-Aware Feature-Wise Linear Modulation.**
Rongyu Zhang, **Yulin Luo**, Jiaming Liu, Huanrui Yang, et al., Shanghang Zhang.
AAAI Conference on Artificial Intelligence (AAAI), 2024.

RESEARCH INTERNSHIPS

Simplexity Robotics, Research Intern Dec. 2025 – Present

- Research on embodied intelligence and generalizable robot foundation models.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern Aug. 2024 – Oct. 2025

- Research on embodied multimodal foundation models and robot intelligence.

ByteDance, Research Intern Mar. 2024 – Aug. 2024

- Trust & Safety Applied Algorithm Team for Ads and E-commerce.

HONORS AND AWARDS

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|---|-----------|
| • Shanghai Jiao Tong University Outstanding Graduate | Jun. 2023 |
| • Shanghai Jiao Tong University B-Class Merit Scholarship | Dec. 2022 |
| • Weichai Power Scholarship | Oct. 2022 |
| • Huawei Scholarship | Dec. 2021 |
| • Shanghai Jiao Tong University B-Class Merit Scholarship | Dec. 2021 |
| • Shanghai Jiao Tong University B-Class Merit Scholarship | Dec. 2020 |

ACADEMIC SERVICE

- Reviewer for CVPR, NeurIPS, ICLR, AAAI, and ECCV.

SKILLS

- Programming: Python, C++, MATLAB.
- Tools: PyTorch, $\text{\LaTeX}$, Git, foundation-model evaluation and data pipelines.